

Homework 5

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function for which we want to find a local minimum using the following *gradient descent* method: define

$$x_{n+1} = x_n - \eta f'(x_n),$$

where $x_1 \in \mathbb{R}$ is a starting value and $\eta > 0$ is the step size (also called learning rate).

Notice that if $f'(x_n) > 0$, then $x_{n+1} < x_n$; if $f'(x_n) < 0$, then $x_{n+1} > x_n$; if $f'(x_n) = 0$, then $x_{n+1} = x_n$ and the algorithm would end. If the sequence (x_n) has a limit x , then $f'(x) = 0$.

Using `Python` explore the following:

- (a) Take a particular convex function f and show that for a small value of η the method converges to the minimum of f .
- (b) Show that by using a larger η the method can converge faster (in fewer steps).
- (c) Show that taking η too large might lead to the divergence of the method.
- (d) Take a particular nonconvex function f and show that, depending on the starting value, the method can end up in a local minimum, not in the global minimum.